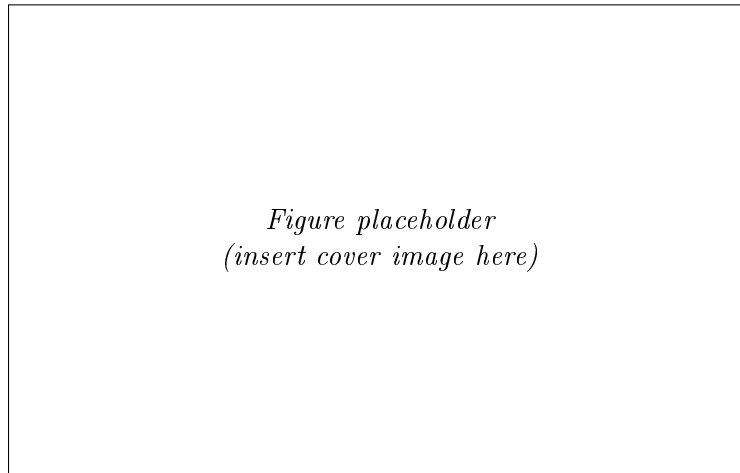


Technical Design Document

Cubli



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Space Challenges 2026

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1 Introduction

1.1 Background & Problem Statement

The Cubli is a reaction-wheel inverted pendulum: a cube-shaped platform that balances on an edge or a corner using only internal actuation, with no external supports or contact forces beyond the single line or point of contact with the ground. The concept was introduced by the Institute for Dynamic Systems and Control (IDSC) at ETH Zurich [1, 2], whose original prototype ($15 \times 15 \times 15$ cm) mounts one brushless DC motor and flywheel on each of three orthogonal faces. Commanding a change in wheel speed produces, by conservation of angular momentum, an equal and opposite reaction torque on the cube body; braking a wheel abruptly transfers its stored angular momentum to the body in a near-impulsive event, which is what allows the cube to *jump up* from resting on a face into a balance on an edge or corner.

Dynamically, the system is an *underactuated*, nonlinear, unstable-equilibrium control problem: three actuators (the wheels) must stabilize a body with more degrees of freedom than actuated inputs, about equilibria (edge and corner balance) that are open-loop unstable and increasingly coupled across axes as the contact geometry shrinks from a line (edge) to a point (corner). This is structurally the same problem as the classical inverted pendulum, extended from a single rotational axis to a fully three-dimensional, momentum-exchange-actuated body — which is why the Cubli has become a standard benchmark platform in the controls community over the past decade.

The problem this project addresses is therefore twofold: (i) derive and validate a dynamic model and controller capable of stabilizing this underactuated system about its unstable equilibria and driving it through commanded reorientations, and (ii) realize this in hardware as a modular, largely parametric design, since final actuator and sensor specifications are not yet confirmed at the time of writing. The target capability set, in order of increasing difficulty, is edge balance, corner balance, controlled rotation, and walking (Section 1.3).

Why this is a hard build, not a decorative demo. It is worth stating plainly why a machine that looks like a cube with three flywheels resists casual replication. The difficulty is not that any single subsystem is exotic — none is — but that every subsystem sits close to a physical bound, and the bounds are *coupled*: relieving one tightens another. The following failure modes recur across published and amateur replications, and each is a consequence of that coupling rather than of poor workmanship.

- **The jump-up is an energy problem with no margin.** The wheel must store, at its maximum speed, enough angular momentum to lift the cube’s center of mass over the corner. The required per-wheel inertia scales as $ML^{3/2}$ (Section 4.3.1), so mass added *anywhere* — battery, frame, fasteners — demands a larger wheel, and the wheel is itself part of M . The sizing problem is therefore a fixed-point iteration rather than a one-pass calculation, and it does not always converge: for the present cube the three-wheel set alone accounts for roughly a quarter of the total mass budget, and shrinking the wheel diameter to save space makes the situation worse, not better, because inertia must then be recovered with ballast at a smaller radius.
- **Braking torque fights the structure.** The momentum accumulated over seconds of spin-up must be shed in tens of milliseconds. That impulse passes through one motor shaft, one hub, and one motor mount — the very parts the mass budget wants thin. Many builds fail here first, and they fail in a characteristic way: the electronics are fine, the controller is sound, and the coupling, hub or mount yields on the first *successful* jump. The structural load case is set by the maneuver that proves the machine works.

- **Sensing, not the control law, sets the balancing limit.** The attitude estimate degrades from two directions at once: MEMS gyroscope bias drifts, and the accelerometer — the only absolute reference for tilt — is corrupted by vibration that the controller itself generates by spinning the wheels. Structural design and state estimation are consequently not separable concerns. A wheel half a gram out of balance at rim radius can render a perfectly correct estimator unusable, which is why wheel balance is treated here as an estimation requirement and not merely as a mechanical nicety.
- **The plant is genuinely nonlinear and constrained.** Corner balancing does not decouple into three independent single-axis problems: the axes are kinematically coupled through the contact point and the body inertia tensor. The actuators saturate in *both* torque and speed, and under any persistent disturbance the wheels drift toward their speed limit, so stored momentum must be actively managed rather than left to accumulate. Linearizing about the upright equilibrium is entirely adequate for a balancing demonstration and inadequate for anything more interesting [3].

The common thread is worth making explicit, because it determines how this project is sequenced. Replications typically fail not because the control law was wrong, but because the machine was never capable of the maneuver being asked of it: insufficient wheel inertia, motors that cannot reach their rated speed under load, a brake too slow to approximate an impulse, or a mass budget that quietly grew past the point where the stored energy suffices. Every one of these is decidable on paper, before any material is cut — and commonly is not decided there. That is the reason sizing is treated in this document as the first engineering activity rather than a detail to be settled once parts arrive.

1.2 Purpose, Application & Mission Context

The Cubli is not presented here as a balancing novelty but as a terrestrial, gravity-loaded demonstrator of **momentum-exchange attitude control** — the same physical mechanism that governs how real spacecraft point themselves. Nearly every three-axis-stabilized satellite holds and changes its attitude using reaction wheels: flywheels whose commanded speed change produces a reaction torque on the spacecraft body, exactly as in the Cubli. Spacecraft typically carry three or four such wheels (often in a pyramidal arrangement for redundancy) — directly analogous to the Cubli’s three orthogonal wheels — and this reaction-wheel approach is the standard for CubeSats and small satellites in particular, making the Cubli representative of a real, currently-flown hardware class rather than an idealized toy problem. Reaction wheels are distinct from control moment gyros (CMGs), which gimbal a constantly-spinning wheel instead of changing its speed to produce much larger torques [4]; positioning the Cubli explicitly as a *reaction-wheel*, not CMG, demonstrator keeps the application claim precise.

Framed as an ADCS problem, the Cubli’s core behaviors map directly onto a spacecraft’s day-to-day pointing task:

- **Edge / corner balance** → regulation to a commanded reference attitude and disturbance rejection — holding a pointing setpoint against perturbations.
- **Controlled rotation** → a slew maneuver: reference tracking to a new commanded attitude, exactly as a satellite re-aims a payload or antenna.
- **Jump / walk** → momentum-exchange hopping mobility, flight-proven on small bodies where wheeled traction is impossible. JAXA’s Hayabusa2 mission deployed the MINERVA-II1 rovers [5] and the MASCOT lander [6] on asteroid Ryugu (2018), both of which move by spinning up and braking an internal flywheel or eccentric mass to hop across the surface, with MASCOT additionally self-righting the same way. Walking, as a sequence of

controlled, directional hops, is a genuine step beyond a single ballistic hop and is an active research frontier (e.g. ETH's SpaceHopper [7]).

This gives the project two complementary application narratives rather than one: it is simultaneously a low-cost, hardware-in-the-loop testbed for standard spacecraft *attitude determination and control* (ADCS), and a ground analog for the momentum-exchange *surface mobility* now used on small-body missions. Framed against a companion star-tracker project, which answers “*where am I pointing?*” (attitude determination), the Cubli answers “*drive me to that reference and hold it*” (attitude control) — the two halves of a complete ADCS loop.

Why the problem is still open. The nominal Cubli problem — balance on a corner, jump up — is solved, and was solved a decade ago on well-instrumented, precisely manufactured, generously funded hardware [1, 2, 3]. Reproducing that result is an engineering exercise, not a research contribution, and this document does not claim otherwise. The interesting questions begin immediately *after* that point, and they are not Cubli-specific: they are the questions a spacecraft attitude-control engineer faces, in a form small enough to be answered on a bench.

- **Control under simultaneous torque and momentum saturation.** A jump saturates the wheels by construction — the maneuver is defined by running them to their speed limit and dumping the momentum. Steering a nonlinear attitude system to a target while respecting a finite momentum envelope, and desaturating without losing the pointing objective, is still handled largely case-by-case with tuned heuristics rather than by a general method.
- **Recovery from large attitude excursions.** Linear controllers valid near upright say nothing useful once the body is far from equilibrium, which is precisely the regime a tumbling or mis-deployed spacecraft occupies. Nonlinear and predictive methods make claims there, and verifying those claims empirically requires a system in which a failed recovery costs a re-print rather than a mission.
- **Robustness to badly known parameters.** Real inertia tensors, friction coefficients and center-of-mass offsets are known to a few percent at best, and shift with assembly, thermal state and wear. Adaptive and robust designs are overwhelmingly validated against *simulated* parameter errors, which are drawn from the same distribution the designer assumed.
- **Whether high-performance methods survive commodity sensing.** Attitude-control techniques developed against clean measurements must eventually run on a MEMS IMU with drifting bias and vibration-corrupted accelerometers. Whether a given method degrades gracefully or collapses under that sensing is an empirical question, and it is settled by hardware rather than by analysis.

Why a Cubli, and why built from scrap. The Cubli is the cheapest honest test of the four questions above. Simulation cannot answer them, because the effects that decide the outcome are exactly the ones a model omits: unmodeled friction, frame resonance excited by the wheels, and the interaction between wheel imbalance and estimator bias. These are what separate a controller that works in simulation from one that works on hardware. At the other extreme, a full-scale testbed — air bearing, vacuum chamber, flight-grade wheels — costs orders of magnitude more and, decisively, *cannot be crashed repeatedly*. The scrap-built cube occupies the useful middle: physical enough to be honest, cheap enough to be abused. The choice to build it from commodity and salvaged parts is therefore methodological rather than merely budgetary, on four counts:

- **Cheap failure means the aggressive experiments actually get run.** A controller that must never fail cannot be tested at its limits, and a testbed too valuable to break silently biases the experimental program toward maneuvers already known to work.

- **Cheap parts are a stronger test than good parts.** Loose tolerances, an imperfectly balanced wheel and a drifting IMU are not defects to be apologized for — they *are* the parameter uncertainty and disturbance content that robust and adaptive methods claim to tolerate. A demonstration on deliberately imperfect hardware therefore carries evidence that a demonstration on precision hardware does not.
- **Constraints force honest sizing.** With a fixed motor, a fixed print volume and a fixed budget, a marginal design cannot be rescued by specifying a larger component. This is the entire justification for the hex-nut-in-pocket ballast scheme of Section 5.3.3: it buys inertia at the outer radius using standard steel nuts and no machined part, and whether it closes is a calculation that must be completed *before* printing, not discovered afterwards.
- **A commodity bill of materials is reproducible.** Every part is orderable or salvageable by any university group, so the result can be checked by others — which is the difference between a demonstration and a claim.

Mission context. The subsystem exercised here is the one a spacecraft uses for pointing: momentum-exchange attitude control under actuator saturation, with desaturation, on an underactuated body whose inertia is imperfectly known. In one respect the terrestrial case is strictly *harder* than the orbital one. In orbit, a marginal attitude controller degrades slowly and often invisibly — pointing error grows, wheels creep toward saturation, and the fault may take orbits to become obvious. Under gravity, the destabilizing torque never vanishes and never relents: an inadequate controller falls over immediately and unambiguously. The demonstration is, in that sense, self-evaluating, and this is a feature of the platform rather than a limitation of it.

1.3 Objectives

The project targets four behaviors, deliberately ordered as a difficulty ramp in which each milestone is a control precondition for the next:

1. **Edge balance** — stabilize the cube on a single edge, the least unstable equilibrium (contact is a line, one primary axis of instability).
2. **Corner balance** — stabilize the cube on a single vertex, the fully unstable equilibrium (point contact, all three axes coupled and unstable simultaneously).
3. **Controlled rotation** — execute a commanded slew between equilibria (e.g. edge to edge, or in-place reorientation), demonstrating reference tracking rather than pure regulation.
4. **Walking** — chain repeated, directional controlled falls between edges/corners into a net translation across a surface, re-stabilizing after each transition.

These map one-to-one onto the applications of Section 1.2: (1)–(2) demonstrate pointing regulation and disturbance rejection of increasing difficulty; (3) demonstrates slew/reference-tracking; (4) demonstrates momentum-exchange mobility of the kind flown on Hayabusa2. Milestones (1)–(3) are treated as committed deliverables for this design; walking (4) is scoped as the stretch objective, contingent on hardware confirmation and the jump-up feasibility check (angular momentum required to tip the cube vs. available motor/wheel torque), to be resolved once final components are known.

2 Project Organization

2.1 Team Organization

The project is developed by a multidisciplinary team, with responsibilities distributed across the main technical domains of the Cubli: mathematical modeling, control, mechanical/CAD design, and system integration. Table 1 summarizes each member’s primary role, and the detailed responsibilities are outlined below.

Table 1: Team members, contact details and primary roles.

Member	Contact	Primary Role
Pablo Urioste Alarcón	pb.urioste4@gmail.com +34 654 491 174	System integration and lead: digital actuator and sensor coding, embedded software, and integration of the overall design.
Deyan Nikolaev Vlaev	vlaev.dido@gmail.com +359 877 082 499	Mathematical demonstration and derivation of the system dynamics.
Niccolo	—	Control system design and analysis.
Andrea Liang	andrea.liang.2006@gmail.com +31 06 5796 2976	Support on the control problem and structural/mechanical integration.
Suvanna Viriyanti Wu	Viriyanti.wu@gmail.com +39 329 074 7550	CAD modeling and mechanical design.
Athanasia Nikolova	—	CAD modeling, prototype building/assembly, and product presentation (commercial pitch).

Responsibilities.

- **Pablo Urioste Alarcón** — System integration and overall coordination. Handles the coding of the digital actuators and sensors, the embedded software stack, and the integration of the mechanical, electronic and control subsystems into a working prototype.
- **Deyan Nikolaev Vlaev** — Mathematical foundation. Responsible for the analytical demonstration and derivation of the equations of motion and the dynamic model of the Cubli.
- **Niccolo** — Control. Leads the design, tuning and analysis of the control strategy used to balance and maneuver the Cubli.
- **Andrea Liang** — Control and mechanics support. Assists on the control problem and contributes to the structural and mechanical integration of the system.
- **Suvanna Viriyanti Wu** — CAD and mechanical design. Develops the 3D models and mechanical drawings of the structure and reaction-wheel assembly.
- **Athanasia Nikolova** — CAD, prototype building and commercialization. Contributes to the 3D modeling, leads the physical building and assembly of the prototype, and is responsible for presenting and pitching the product.

2.2 Work Breakdown Structure

This plan runs from project start to the final presentation on 20 August 2026. The documentation stream is front-loaded and closes with the Technical Design Document deliverable on 3 August,

while the control and hardware streams continue through the build, balancing and jump-up demonstrations up to the presentation. The work is grouped into three streams — investigation, requirements and documentation; control and software; and hardware design and build — led by the owners assigned in Table 1. The investigation is kept to the essentials needed to substantiate the design: a focused literature review, the dynamics derivation, and the wheel sizing that follows from it. The table below breaks the three streams into individual tasks, each with its scope, owner and dates.

ID	Task	Scope	Owner	Dates
Investigation, Requirements & Documentation (to the 3 Aug TDD deliverable)				
A1	Literature & state of the art	Review the ETH Cubli, the one-wheel Cubli, CubeSat ADCS practice and the hopping-mission precedents	Suvanna (+Dejan)	23–25 Jul
A2	Problem statement & framing	Lock the ADCS application narrative that the rest of the document refers back to	Dejan	23–24 Jul
A3	Scope & requirements	Define scope and build the requirements table (functional, performance, design constraint) with verification methods	Dejan (+Pablo)	24–25 Jul
A4	Milestones & roles	Prototype staging, binary gate criteria, and a one-page RACI of owners	Dejan	23–25 Jul
A5	Risk register	Trigger, impact, owner and mitigation for the leading risks	All (+Pablo)	28–29 Jul
A6	1-DoF equations of motion	Lagrangian derivation of the reaction-wheel pendulum with friction terms	Dejan (+Andrea)	24–25 Jul
A7	3D equations of motion	Coupled three-wheel dynamics for the edge and corner pivots	Dejan (+Andrea)	27–28 Jul
A8	Linearisation & controllability	Equilibria, linear models, controllability/observability checks	Dejan (+Andrea)	29 Jul
A9	Jump-up sizing & budget	Energy-method wheel sizing and the parametric mass/inertia/CoM budget	Dejan (+Andrea)	30 Jul–1 Aug
A10	TDD draft (all sections)	Every member drafts their section; placeholder figures acceptable	All	25–26 Jul
A11	TDD deepen + S1 data	Full modelling and control sections; insert the rig evidence and measured parameters	All	1–2 Aug
A12	TDD assemble & submit	Consistency pass, figures, references, submission of the deliverable	All (Dejan integrates)	3 Aug
Control & Software				
B1	Sim environment + plant	Simulation setup and the 1-DoF plant, validated by energy conservation	Andrea	23–24 Jul
B2	LQR design (1-DoF, sim)	Tune the regulator and sweep robustness to inertia error, delay and saturation	Andrea (+Dejan)	26–27 Jul
B3	State estimator	Complementary tilt filter with injected noise and bias in simulation	Nicc	27–28 Jul
B4	Firmware & comms	Firmware architecture, loop rates, and the CAN-FD command/reply layer	Nicc (+Pablo)	24–28 Jul
B5	Sensor drivers + cal	IMU and encoder drivers, gyro/accel calibration and the lever-arm correction	Nicc (+Andrea)	28–30 Jul
B6	Safety & mode logic	Watchdog, tilt-limit disarm, wheel-speed cap and the mode state machine	Nicc (+Andrea)	30–31 Jul

ID	Task	Scope	Owner	Dates
B7	S1 bring-up & balance	Port estimator and LQR to the rig and tune to a stable balance	Nicc (+Andrea)	31 Jul–2 Aug
B8	System identification	Measure K_t , wheel inertia, friction and loop latency on the rig	Andrea (+Nicc)	1–2 Aug
B9	3D model + LQR (sim)	3D plant and the edge/corner regulators with the CAD inertia tensor	Andrea (+Dejan)	3–6 Aug
B10	EKF / MEKF estimator	3D estimator with gyro-bias states; characterise the yaw drift	Nicc (+Andrea)	6–8 Aug
B11	Edge balance (S2a)	Retune the edge controller on the assembled cube; log disturbance recovery	Andrea (+Nicc)	10–13 Aug
B12	Corner balance + slew (S2b/c)	Coupled three-axis corner balance and a commanded slew; report ADCS metrics	Andrea (+Nicc)	14–16 Aug
B13	Jump-up (S3)	Spin-up profile and brake release for a face-to-edge jump	Nicc (+Andrea)	17–18 Aug
Hardware Design & Build				
C1	Envelope & layout study	Packaging of wheels, drivers, battery and electronics with a central CoM	Neisa (+Suvanna)	23–24 Jul
C2	Reaction-wheel design	Wheel to the inertia target: radius, rim, hub bolting to the motor bell	Suvanna (+Neisa)	24–26 Jul
C3	Mounts & brackets	Motor mount, ring-magnet carrier and adjustable encoder bracket	Neisa, Suvanna	27–29 Jul
C4	1-DoF rig CAD + jigs	Single-axis rig and the balancing/assembly tooling	Suvanna (+Neisa)	25–28 Jul
C5	Mass properties & freeze	Export the inertia tensor/CoM and release the manufacturing package	Neisa (+Suvanna)	2–3 Aug
C6	Procurement order	Order the long-lead items (rims, frame stock, connectors, spares)	Pablo (+Neisa)	23–24 Jul
C7	Bench PSU & bring-up	Current-limited supply, safety kit, and single-axis electrical bring-up	Pablo (+Nicc)	25–27 Jul
C8	Fabricate wheels (×4)	Cut/receive rims, print hubs, bond magnets — three plus one rig spare	Neisa, Suvanna	27–30 Jul
C9	Balance & spin-test	Static-balance and containment spin-test every wheel	Suvanna (+Neisa)	30–31 Jul
C10	Assemble 1-DoF rig	Motor, balanced wheel, encoder gap, IMU and hard stops	Pablo (+Neisa)	31 Jul–1 Aug
C11	Print structure + frame	Structural print set and the fabricated, orthogonal frame	Neisa, Suvanna	3–6 Aug
C12	Perfboard + harness	Power/signal board and the full power and CAN harness	Pablo	3–6 Aug
C13	Cube integration + power-on	Assemble the three axes, then bench and battery smoke test	Pablo (+Neisa)	7–9 Aug
C14	Brake mechanism build	Fabricate and fit the servo brake; bench-test the deceleration	Suvanna (+Pablo)	10–13 Aug
Deliverables				
D1	Results, demo & final presentation	Slow-motion capture, results plots, deck, and the final presentation	All (+Suvanna edits)	17–20 Aug

2.3 Schedule and Contingency Plan

The full-page schedule in Figure 1 places every task of Section 2.2 on the 23 July–20 August 2026 calendar, and Table 3 states the gate that closes each phase. The documentation stream is deliberately completed early, at the TDD deliverable on 3 August, so the remaining weeks are free for the build, balancing and jump-up work that culminates in the final presentation. The critical path runs through the dynamics investigation and wheel sizing, which feed both the control and the hardware streams; the single-axis (S1) rig is the protected item, since its balance result is the experimental evidence in the document. If any modeling or rig task slips, the TDD is still submitted on schedule with simulation-only evidence and the affected result carried into the later build weeks.

Table 3: Milestone gates, from kick-off to the final presentation.

Gate	Pass criterion	Date
G1	Long-lead order placed; requirements and plan locked	24 Jul
G2	Dynamics, control design and wheel sizing complete	1 Aug
G3	1-DoF rig balances ≥ 60 s; plant parameters measured	2 Aug
G4	TDD delivered (documentation deliverable)	3 Aug
G5	Cube integrated and powered; all axes addressable	9 Aug
G6	Edge balance (S2a) for ≥ 60 s with disturbance recovery	13 Aug
G7	Corner balance and commanded slew (S2b/c)	16 Aug
G8	Face-to-edge jump-up (S3) captured on video	18 Aug
G9	Final presentation and demonstration	20 Aug

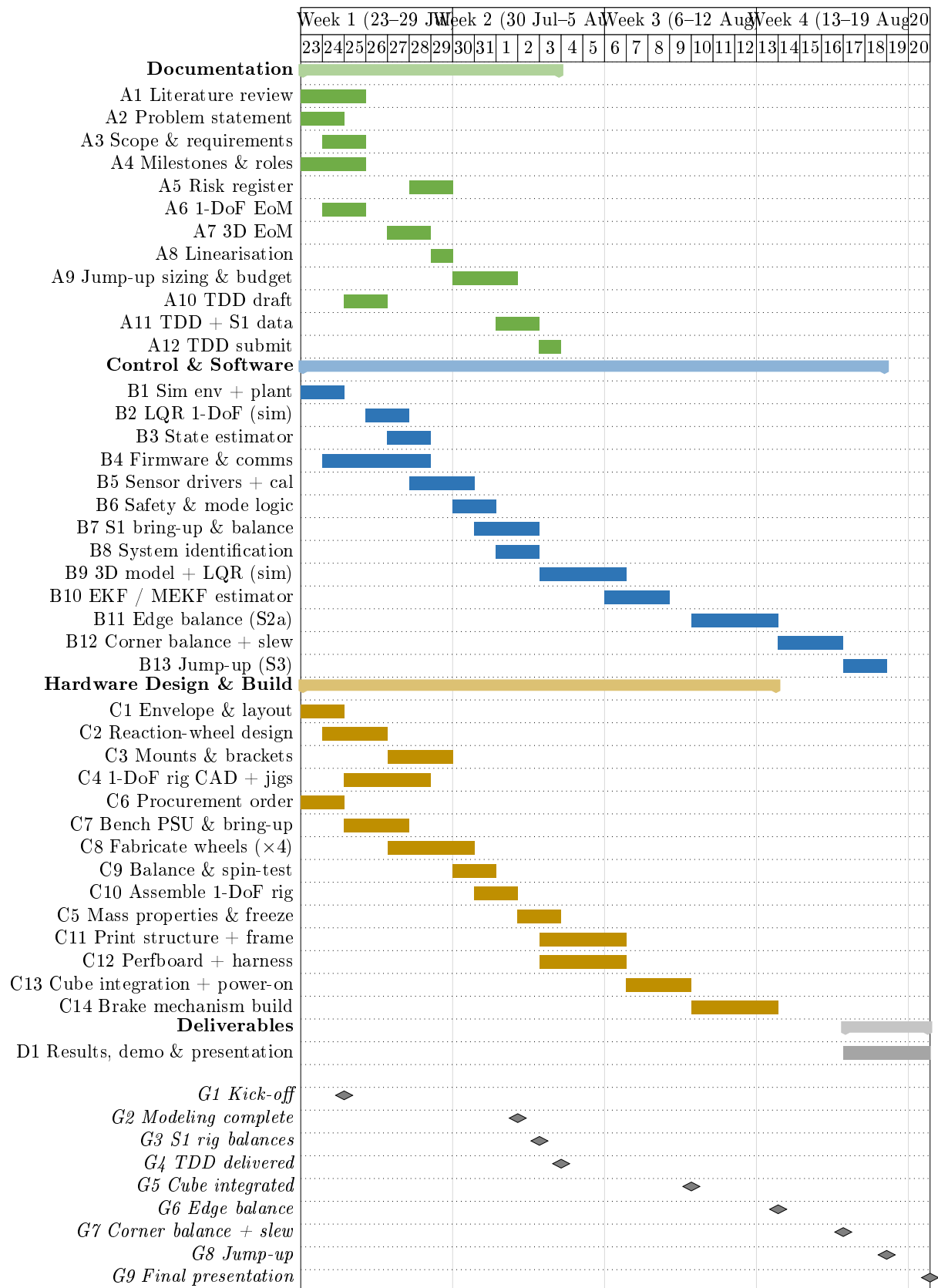


Figure 1: Detailed project schedule, 23 July–20 August 2026. The documentation stream (green) closes at the TDD deliverable on 3 August; control (blue) and hardware (gold) continue through the build and demonstrations to the final presentation on 20 August. Grey diamonds: milestone gates.

3 System Analysis & Requirements

3.1 Scope of Work & Use Cases

3.2 Actuation Trade-off — Why Reaction Wheels

3.3 Requirements & Objectives Traceability

Each project objective (Section 1.3) imposes a required capability, which in turn is confirmed by a specific test in the validation plan (Section 6). The mapping below keeps every requirement traceable from motivation to verification.

Table 4: Objective-to-requirement-to-verification traceability.

Objective	Required capability	Verified by
Edge balance	Regulation about a line-contact equilibrium	Sec. 6
Corner balance	Regulation about a point-contact equilibrium, all axes	Sec. 6
Controlled rotation	Reference tracking / slew between equilibria	Sec. 6
Walking (stretch)	Chained jump-up + re-stabilization	Sec. 6

3.4 Coordinate Frames & Conventions

4 Dynamic Modeling & Control

The full Cubli is underactuated and simultaneously unstable about all three axes when balanced on a corner (Section 1.3). The modeling, identification, and control workflow is therefore first validated on a two-dimensional (2D) single-face prototype with one rotational degree of freedom, before being generalized to the 3D cube in Section 4.5.

4.1 2D Reaction-Wheel Pendulum

The 2D prototype is a 3D-printed plate sized to one cube face, with a motor-driven momentum wheel at its center and a braking mechanism at one corner. The plate pivots on a bearing at that corner, constraining motion to a single rotational degree of freedom in a vertical plane.

4.1.1 Equations of Motion

The nonlinear equations of motion are

$$\ddot{\theta}_b = \frac{(m_b l_b + m_w l) g \sin \theta_b - T_m - C_b \dot{\theta}_b + C_w \dot{\theta}_w}{I_b + m_w l^2}, \quad (1)$$

$$\begin{aligned} \ddot{\theta}_w = & \frac{(I_b + I_w + m_w l^2)(T_m - C_w \dot{\theta}_w)}{I_w(I_b + m_w l^2)} \\ & - \frac{(m_b l_b + m_w l) g \sin \theta_b - C_b \dot{\theta}_b}{I_b + m_w l^2}, \end{aligned} \quad (2)$$

with motor current-torque relation $T_m = K_m u$. Symbols are defined in Table 5 and used throughout this section.

Table 5: 2D pendulum model variables.

Symbol	Description	Unit
θ_b	Body tilt angle	rad
θ_w	Wheel rotation relative to body	rad
m_b, m_w	Body, wheel mass	kg
I_b, I_w	Body (about pivot), wheel (about motor axis) inertia	kg m ²
l	Motor axis to pivot distance	m
l_b	Body CoM to pivot distance	m
g	Gravitational acceleration	m/s ²
T_m	Motor torque	N m
C_b, C_w	Body, wheel damping coefficient	kg m ² /s
K_m	Motor torque constant	N m/A
u	Motor current input	A

4.1.2 Equilibria & Linearization

The relevant equilibrium is the upright position $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$; its linearization is presented with the control design in Section 4.4.

4.2 Parameter Identification & the 2D Testbed

CAD mass-properties simulation, accounting for print material, infill, and mounted hardware (driver, encoder, MCU, gyroscope/accelerometer, wiring), gives m_b, m_w, I_b, I_w directly, while l and l_b follow from the plate's design geometry. The damping coefficients C_b and C_w cannot be obtained from CAD, since they depend on bearing friction, wiring drag, and motor cogging; nominal values will be estimated for the initial control design, with dedicated testing planned once the prototype is fabricated.

Table 6: Parameter determination method (symbols per Table 5).

Symbol	Determination method
l	Calculated during design
l_b	Calculated during design
m_b	CAD mass properties
m_w	CAD mass properties
I_b	CAD mass properties
I_w	CAD mass properties
C_b	Estimated (future testing planned with prototype)
C_w	Estimated (future testing planned with prototype)
K_m	Motor datasheet

4.3 Jump-Up Feasibility Analysis

Given CAD-derived $I_b, I_w, m_b, l_b, m_w, l$, the wheel speed ω_w required for jump-up is estimated before fabrication, assuming a perfectly inelastic wheel-body collision. Conservation of angular momentum at impact and of energy from impact to edge-standing yield

$$\omega_w^2 = (2 - \sqrt{2}) \frac{I_w + I_b + m_w l^2}{I_w^2} (m_b l_b + m_w l) g. \quad (3)$$

This value is checked against the motor's achievable speed at the available bus voltage and used to size wheel inertia (radius, rim mass) within the motor's operating range without excessively reducing the balancing recovery angle. Post-fabrication, ω_w is refined experimentally to correct for deviations from the ideal collision assumption.

The braking mechanism uses a servo-driven barrier that strikes a bolt head on the wheel to stop it abruptly. Servo response time bounds the validity of the “instantaneous stop” assumption; a slower stop yields a lower effective post-impact body velocity than predicted, compensated through the same tuning process.

4.3.1 Torque-Limited Wheel-Inertia Target

The relation above assumes an instantaneous, lossless momentum transfer. For preliminary sizing, the achievable motor speed is treated as fixed and the required per-wheel inertia is derived from a momentum budget that additionally accounts for finite brake torque. This gives an ideal per-wheel angular momentum

$$h_{w,\text{ideal}} = \sqrt{\eta} \frac{\sqrt{2g\lambda\kappa}}{n} ML^{3/2}, \quad (4)$$

where M, L, g are cube mass, edge length, and gravity, η is an energy margin, and κ, λ, n are dimensionless geometric factors set by the contact case (Table 7).

The corner case is more demanding and governs the design.

Table 7: Contact-case geometric factors.

Case	κ (pivot inertia)	λ (CoM rise)	n (momentum transfer)
Edge	2/3	$1/\sqrt{2} - 1/2$	1
Corner (governs design)	11/12	$\sqrt{3}/2 - 1/2$	$\sqrt{2}$

A finite brake torque τ_b cannot deliver the impulse instantaneously and must exceed the gravitational tip-over torque $\tau_g = MgL/2$. This loss is captured by a brake-amplification factor

$$\beta = \frac{\tau_b}{\tau_b - \tau_g}, \quad h_w = \beta h_{w,\text{ideal}}, \quad I_{w,\text{target}} = \frac{h_w}{\omega_{\max}}, \quad (5)$$

where $\omega_{\max}$ is the maximum practical wheel speed. Table 8 gives the resulting target for the baseline cube, which the reaction-wheel design in Section 5.3.3 must meet.

Table 8: Torque-limited wheel-inertia target (corner case, baseline cube).

Symbol	Description	Value
M	Cube mass	1.45 kg
L	Cube edge length	180 mm
η	Energy margin	1.05
τ_b	Brake torque per wheel	5.0 N m
τ_g	Gravitational tip-over floor	1.28 N m
β	Brake-amplification factor	1.34
$\omega_{\max}$	Max wheel speed (design)	6000 rpm (628 rad/s)
h_w	Required angular momentum	0.277 kg m ² /s
$I_{w,\text{target}}$	Target wheel inertia	4.40×10^{-4} kg m²

Choice of $\omega_{\max}$. The maximum wheel speed is a design choice rather than a datasheet figure, and it enters the sizing directly through $I_{w,\text{target}} = h_w/\omega_{\max}$. At $K_v = 380$ rpm/V on a nominal 22.2 V pack the motor's no-load speed is 8436 rpm, but this is unattainable in service: winding resistance (2×194 m Ω line-to-line), the reduced phase voltage available under sinusoidal field-oriented modulation, and the fall of pack voltage through the discharge curve together place the realistic ceiling near 6900 rpm. The design value is set to **6000 rpm (628 rad/s)**, below even that figure. At this speed the back-EMF is 15.8 V, only 71 % of the nominal bus, so the wheel reaches its commanded speed across the whole usable discharge range rather than only on a freshly charged pack, and margin remains for motor tolerance, bearing and aerodynamic drag, and the residual uncertainty in the mass budget. The cost is a 40 % larger inertia requirement than a top-of-range assumption would give, which the wheel of Section 5.3.3 meets with 2.4 % margin. The corresponding electrical frequency, 1200 Hz at 12 pole pairs, sits well inside the driver's capability.

4.4 Control Design on the 2D Model

4.4.1 LQR Baseline Control

Linearizing about the upright equilibrium $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$ gives

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x := (\theta_b, \dot{\theta}_b, \dot{\theta}_w), \quad (6)$$

with A, B built from the parameters in Table 5. Zero-order-hold discretization at a rate set by the open-loop unstable pole, with a safety margin, gives $x[k+1] = A_d x[k] + B_d u[k]$. A discrete LQR gain $K_d = (K_{d1}, K_{d2}, K_{d3})$ minimizes

$$J(u) = \sum_k x^T[k] Q x[k] + u^T[k] R u[k], \quad (7)$$

giving $u[k] = -K_d(\hat{\theta}_b[k], \hat{\theta}_b[k], \hat{\theta}_w[k])$.

The tilt estimate $\hat{\theta}_b$ is obtained from a gyroscope and accelerometer mounted at, or as close as mechanically feasible to, the pivot point rather than at the plate's center. This placement is necessary because an accelerometer at radius r from the pivot measures gravity plus the tangential ($r\ddot{\theta}_b$) and centripetal ($r\dot{\theta}_b^2$) acceleration components induced by the plate's rotation, which corrupt the naive estimate $\tan^{-1}(-a_x/a_y)$ during fast motion (e.g., jump-up or aggressive recovery); placing the sensor at $r \approx 0$ makes both terms negligible. Tilt is estimated with a complementary filter combining gyro integration (drift-prone, unaffected by the terms above) with accelerometer inclination (drift-free, but unreliable during shocks):

$$\hat{\theta}_b[k] = (1 - \gamma)(\hat{\theta}_b[k-1] + \dot{\theta}_{b,\text{gyro}}[k] \Delta t) + \gamma \theta_{b,\text{accel}}[k], \quad (8)$$

where $\theta_{b,\text{accel}}[k] = \tan^{-1}(-a_x[k]/a_y[k])$ is the instantaneous accelerometer-only estimate and $\gamma \in (0, 1)$ is a small weight favoring the gyro term between corrections.

A constant tilt-sensor offset d , including residual bias from imperfect pivot placement or sensor misalignment, would otherwise appear as a nonzero steady-state wheel speed rather than a corrected angle. A low-pass filter state

$$x_f[k+1] = (1 - \alpha)x_f[k] + \alpha\hat{\theta}_b[k] \quad (9)$$

is subtracted from the tilt feedback,

$$u[k] = -K_d(\hat{\theta}_b[k] - x_f[k], \hat{\theta}_b[k], \hat{\theta}_w[k]), \quad (10)$$

so d is absorbed by x_f at steady state, requiring no separate calibration stage. This fixed-gain design is the baseline balancing controller; Table 9 summarizes the control variables.

4.4.2 Gain Scheduling & Mode Transitions

The controller switches from the open-loop jump-up sequence to the LQR law once $|\hat{\theta}_b| < \theta_{\text{th}}$, with hysteresis around θ_{th} to prevent mode chattering. The baseline uses a single fixed gain K_d at the upright equilibrium. If recovery degrades near the edges of the operating range, where the small-angle linearization weakens, gain scheduling will be added: additional gains $K_d^{(i)}$ computed at several linearization points, with the controller interpolating between them based on $\hat{\theta}_b$, reusing the same discrete structure and offset filter.

4.5 Generalization to the 3D Cube

The dynamics, identification workflow, and control approach validated on the 2D model extend to the full three-dimensional cube, where three orthogonal reaction wheels must stabilize the body about its edge and corner equilibria.

4.5.1 Full-Body Equations of Motion

With $R \in SO(3)$ the cube's orientation and $\omega_b \in \mathbb{R}^3$ its body-frame angular velocity, the attitude kinematics are

$$\dot{R} = R[\omega_b]_{\times}, \quad (11)$$

Table 9: Control-design variables (2D and corner-balancing 3D model).

Symbol	Description
x	State vector
A, B	Continuous-time state/input matrices (linearized model)
A_d, B_d	Discrete-time (ZOH) counterparts of A, B
K_d	LQR feedback gain
Q, R	LQR state/input weighting matrices
$J(u)$	LQR cost function
$\hat{\theta}_b, \hat{\dot{\theta}}_b$	Estimated tilt angle, rate (gyroscope/accelerometer)
$\hat{\dot{\theta}}_w$	Estimated wheel rate (encoder)
a_x, a_y	Raw accelerometer readings (tangential, radial axes)
$\theta_{b,\text{accel}}$	Instantaneous accelerometer-only tilt estimate
$\dot{\theta}_{b,\text{gyro}}$	Raw gyro rate measurement
γ	Complementary-filter weight (accelerometer term)
Δt	Control/estimation loop sample period
x_f, α	Offset-correction filter state, filter coefficient
d	Constant tilt-sensor offset
θ_{th}	Tilt threshold, jump-up $\rightarrow$ balancing switch
$K_d^{(i)}$	Scheduled gain at linearization point i (extension)

where $[\cdot]_{\times}$ is the skew-symmetric cross-product matrix. The assembly's angular momentum about the pivot, expressed in the body frame, generalizes the single-axis momentum term of the 2D model:

$$H = I\omega_b + \sum_{i=1}^3 I_{w,i} \omega_{w,i} e_i. \quad (12)$$

Euler's equation gives

$$\dot{H} + \omega_b \times H = \tau_g - C_b \omega_b, \quad (13)$$

and each wheel obeys

$$I_{w,i}(\dot{\omega}_{b,i} + \dot{\omega}_{w,i}) = T_{m,i} - C_{w,i} \omega_{w,i}, \quad i = 1, 2, 3. \quad (14)$$

Table 10 defines all symbols.

Restricting motion to a single vertical plane with two wheel torques set to zero recovers the 2D prototype model exactly, with $I_b, l_b, m_b, m_w, I_w, C_b, C_w$ the corresponding single-axis projections of $I, r_{cm}, m, I_{w,i}, C_{w,i}$.

4.5.2 Edge vs. Corner Equilibria & Linearization

Jump-up proceeds in two stages: Flat-to-Edge, where one wheel brakes, followed by Edge-to-Corner, where a second wheel brakes. **Edge balancing** constrains the body to a single axis, so the dynamics reduce to the 2D form, and the 2D LQR baseline and offset filter apply directly, with m_b, l_b, I_b replaced by their edge equivalents.

Corner balancing is genuinely 3D: all three wheels contribute across three rotational DOF. Linearizing about the corner-up equilibrium $(\theta, \omega_b, \omega_w) = (0, 0, 0)$, with $\theta \in \mathbb{R}^3$ the small orientation error, gives the same state-space form as the 2D case:

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x := (\theta, \omega_b, \omega_w) \in \mathbb{R}^9, \quad (15)$$

Table 10: Full-body (3D) model variables.

Symbol	Description
R	Cube orientation relative to world frame
ω_b	Body angular velocity, body frame ($\in \mathbb{R}^3$)
$\omega_{b,i} = e_i^T \omega_b$	Body angular velocity about wheel axis i
$\omega_w = (\omega_{w,1}, \omega_{w,2}, \omega_{w,3})$	Wheel angular velocities relative to body
e_i	Spin axis of wheel i (face normal)
I	3×3 inertia tensor, body+wheels, about pivot
H	Total angular momentum about pivot
$\tau_g = m r_{cm} \times (R^T \hat{g})$	Gravity torque about pivot
m	Total assembly mass
r_{cm}	Body-frame vector, pivot to assembly CoM
$\hat{g}$	World vertical unit vector
C_b	3×3 body damping matrix
$I_{w,i}, C_{w,i}$	Inertia, damping of wheel i
$T_{m,i} = K_{m,i} u_i$	Motor torque, wheel i
$u_i, K_{m,i}$	Motor current, torque constant, wheel i

with $A \in \mathbb{R}^{9 \times 9}$ and $B \in \mathbb{R}^{9 \times 3}$ built from I , r_{cm} , and the per-wheel parameters $I_{w,i}, C_{w,i}, K_{m,i}$ (Table 10). The LQR design generalizes directly: K_d becomes a 3×9 gain matrix, using the same discretization, cost function, and offset filter as Table 9, with Q, R weighting all three tilt axes, body rates, and wheel speeds. The mode-transition logic gains a third mode, Edge-to-Corner jump-up, triggered once edge balancing is stable, followed by a second threshold switch into the corner LQR once estimated orientation nears corner-up.

5 Preliminary Hardware Design

This design is preliminary: final actuator and sensor specifications are not yet confirmed, so the mechanical and electrical architecture is kept modular and largely parametric (Section 1.1).

5.1 System Architecture

5.2 Mass and Inertia Budget

Before committing to component geometry, the preliminary design is bounded by three coupled constraints that must be satisfied simultaneously. Sizing to the worst case of each fixes the envelope within which the mechanical and electrical design then iterate.

Volume constraint. Every subsystem—the three orthogonal reaction-wheel/motor modules, the battery, and the electronics stack—must fit inside the 180 mm cube envelope. This envelope is not arbitrary. Sizing the wheels to the conservative 6000 rpm of Section 4.3.1 requires 4.40×10^{-4} kg m² per wheel; within a 150 mm envelope the largest wheel that clears the frame, motor and contact features is about 120 mm in diameter, yielding 2.63×10^{-4} kg m²—17% short, and not recoverable by ballast, which necessarily sits inboard of the ring it displaces. The 180 mm envelope admits the 130 mm wheel of Section 5.3.3. Packaging independently demands it: the 150 × 90 mm power and signal board does not fit within a 150 mm cube at any wall thickness. The binding dimension remains the reaction wheel, whose outer diameter caps the radius at which ballast mass can be placed.

Mass constraint. The cube is designed to a total mass target of $M = 1.45$ kg. This is not merely a logistics figure: it feeds back into the dynamics, since the gravitational tip-over torque $\tau_g = MgL/2$ and the required jump-up angular momentum both scale with M (Section 4.3.1). A heavier cube demands a more capable brake and a larger wheel inertia, so the mass target and the inertia target are coupled and must be closed together. The preliminary allocation is given in Table 11, with the reaction-wheel set (the one firmly-sized item) committed and the remaining subsystems held as allowances to be refined as components are selected.

Table 11: Preliminary mass budget (per cube). Reaction wheels sized; others allocated.

Subsystem	Allocated mass (g)	Status
Reaction wheels (3×)	519	Sized (Sec. 5.3.3)
Motors (3×)	TBD	Allowance — pending motor selection
Frame / structure	TBD	Allowance — pending CAD
Battery	TBD	Allowance — pending capacity sizing
Electronics (MCU, drivers, IMU, servo)	TBD	Allowance
Wiring / fasteners / margin	TBD	Allowance
Total (target)	1450	Design target

Inertia constraint (corner-critical). Each reaction wheel must store enough angular momentum at the motor’s maximum speed to execute the jump-up maneuver. The two contact cases are not equally demanding: the corner equilibrium requires a larger rise of the center of mass and a less favorable momentum-transfer geometry than the edge equilibrium, so for a common maximum wheel speed $\omega_{\max}$ it demands the greater wheel inertia. Evaluating the torque-limited budget of Section 4.3.1 for both cases (Table 12), the corner case requires 4.40×10^{-4} kg m² against 3.99×10^{-4} kg m² for the edge —about 10% more. The corner jump-up is therefore the critical condition, and the reaction wheel is sized to it; a wheel that satisfies the corner requirement satisfies the edge requirement with margin.

Table 12: Required per-wheel inertia by contact case, at $\omega_{\max} = 6000$ rpm.

Contact case	h_w (kg m ² /s)	$I_{w,\text{target}}$ (10 ⁻⁴ kg m ²)
Edge	0.251	3.99
Corner (critical)	0.277	4.40

5.3 Mechanical Design

5.3.1 2D Prototype Build

5.3.2 3D Cube Frame

5.3.3 Reaction-Wheel Sizing

The reaction wheel must supply the per-wheel inertia $I_{w,\text{target}} = 4.40 \times 10^{-4}$ kg m² derived from the jump-up budget of Section 4.3.1, while remaining light enough not to dominate the cube mass. Rather than adjusting inertia by scaling a solid disc, the wheel is built as a printed *ring plus spokes* whose inertia is then tuned by *ballast*: standard steel hex nuts dropped into hexagonal pockets distributed around the ring. Each pocket both removes plastic and adds steel, so the net inertia gain per pocket is large and the design target is reached by choosing how many nuts to install—a discrete, easily-adjustable knob that also lets the physical wheel be re-tuned after fabrication to absorb parameter uncertainty.

The structure is printed in PET-CF ($\rho = 1290$ kg/m³): a 130 mm outer-diameter ring of 20 mm radial width and 12 mm thickness, carried by three radial spokes 10 mm wide of the same thickness. The hub and motor interface are neglected in the inertia budget. The ballast is standard M6 steel hex nuts (ISO 4032: 10 mm across flats, 5.2 mm thick, $\rho = 7850$ kg/m³, measured mass 2.5 g), seated in blind hexagonal pockets machined to 5.2 mm depth at the mid-width of the ring, leaving a 6.8 mm plastic floor beneath each nut. Accounting for the plastic each pocket displaces, the *net* contribution per installed nut is +1.87 g of mass and $+5.70 \times 10^{-6}$ kg m² of inertia at the ring radius.

Table 13 sweeps the outer diameter, solving for the number of nuts needed to reach $I_{w,\text{target}}$ and checking that the pockets physically fit. Small wheels need so many nuts that no plastic wall survives between adjacent pockets; the largest diameters reach the target from the bare ring alone, leaving no tuning margin and costing mass.

The selected design is **OD = 130 mm with 18 nuts**: it reaches 4.51×10^{-4} kg m² against the 4.40×10^{-4} kg m² target (102.4%), leaves 7.2 mm of plastic between adjacent pockets and at least 4.0 mm of radial wall in either nut orientation, and keeps the three-wheel set to ~ 485 g. The printed ring supplies 74% of the total inertia and the ballast 23%, so the nuts act as a trim about a structure that nearly meets the target unaided, and the wheel can be adjusted in either direction after fabrication.

Pocket placement. The pockets cannot be spaced uniformly. A spoke occupies $\pm 6.4^\circ$ where it meets the ring and a pocket $\pm 6.3^\circ$ at the ballast radius, so a pocket centre must clear a spoke centre by 12.7° ; with three spokes, no uniform pitch at any usable nut count satisfies this. The eighteen pockets are therefore distributed as six per 120° sector between spokes, spread over the 94.7° of usable arc at a 18.9° pitch. Three-fold symmetry, and hence balance, is preserved, and the arrangement admits at most seven per sector should additional trim be required.

Nut retention. At 6000 rpm each nut experiences 54 N of centrifugal load at the 55 mm ballast radius. The blind-pocket floor carries this in bending, but positive retention is required: a cover

Table 13: Reaction-wheel outer-diameter trade study, at $I_{w,\text{target}} = 4.40 \times 10^{-4} \text{ kg m}^2$ with M6 ballast. N is the nut count reaching the target; the selected design carries 18 for trim margin. Selected design in bold.

OD (mm)	N	Mass (g)	I_w (10^{-4} kg m^2)	I_w/target (%)	3 wheels (g)	Pocket fit
80	221	620.1	4.41	100.3	1860.4	fails — no wall
90	148	449.7	4.41	100.1	1349.1	fails — no wall
100	99	339.2	4.40	100.1	1017.7	fails — no wall
110	64	263.8	4.41	100.3	791.4	fails — no wall
120	37	208.3	4.40	100.1	625.0	fails — no wall
130	18	172.9	4.51	102.4	518.6	OK (7.2 mm)
140	0	139.9	4.51	102.5	419.8	bare ring exceeds target
150	0	152.0	5.73	130.1	456.0	bare ring exceeds target

plate over the pocket mouths is preferred to adhesive, since it preserves the post-fabrication adjustability that motivates the ballast scheme.

5.4 Electrical Design

5.4.1 Components

The electronics and software will be built around a microcontroller communicating over a CAN bus with a motor driver that closes the current loop for the selected brushless motor. Wheel speed will be measured with a magnetic encoder on the motor shaft, and body tilt will be estimated from an onboard IMU. Braking will be actuated by a digital servo driven directly from the microcontroller, and a wireless module will provide a telemetry and debugging link. Power will be supplied by a LiPo battery, stepped down through a DC-DC converter for the logic-level electronics, with the motor driver powered directly from the battery bus.

The full parts list is catalogued in Appendix A (Table 14). The supplied set is an electronics-only BOM; the reaction wheels, the frame and all printed parts are the team’s design responsibility, and their fabrication is the leading schedule risk.

5.4.2 Electrical Diagram

5.5 System Integration

6 Testing and Validation Plan

This plan covers system-level validation. Parameter-identification tests that feed the model directly (swing test, current-step identification) are described with the model they validate in Section [4.2](#).

6.1 Simulation-First Validation

6.2 Bring-up Sequence

6.3 Sensor Testing

6.4 Milestone Acceptance Criteria

6.5 Control Performance Evaluation

7 Future Work

7.1 NMPC & Warm-Start Control (Future Implementation)

7.2 Autonomous Stand-up & Disturbance Rejection

7.3 Single-Wheel Variant / Extensions

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A Bill of Materials

The components supplied for this project constitute an *electronics* bill of materials: the actuation chain, wheel and attitude sensing, on-board compute, communications and power distribution are specified, but no structural or mechanical hardware is. The reaction wheels, the frame, the motor mounts, the fasteners and every printed bracket are the team’s design responsibility, and are treated as such in Section 5. Table 14 lists the supplied items grouped by subsystem. The actuation figures derived from these parts — the torque constant, the achievable torque and wheel speed, and the inertia that sizes the wheel — are developed where they are used, in Sections 4.3.1 and 5.3.3.

The architecture the BOM implies is a two-level control stack: each mjbots moteus-n1 closes a high-rate field-oriented current loop on its motor, using the MA600 encoder for rotor angle, while the Teensy 4.1 runs the outer attitude-estimation and state-feedback loop over CAN-FD — the same inner-tachometer / outer-attitude layering used in spacecraft reaction-wheel assemblies. A mechanical wheel brake, following the original ETH Cubli [2], provides the near-impulsive momentum transfer required for the jump-up.

Procurement is the single largest schedule risk. The long-lead items are not electronic but mechanical: the frame stock and the printed structure gate the entire build, and with three actuation sets and no spare, the loss of a motor or driver in service would force the reduced-actuation one-wheel variant [8]. These items are ordered first. Consumables and bench equipment not covered by the supplied set — wire, connectors, fasteners, filament, the current-limited bench supply and the battery-safety kit — are tracked separately in the build documentation.

Table 14: Supplied bill of materials, grouped by subsystem.

Ref	Item	Qty ^a	Key specifications	Function
Actuation				
1	T-Motor MN4006 KV380	3	18N24P outrunner; 380 rpm/V; 68 g; $\varnothing 44.35$ mm $\times$ 21 mm; 194 m Ω ; peak 16 A, 380 W	Reaction-wheel torque source
2	mjbots moteus- n1	3	10–54 V; 100 A peak / 9 A cont.; 15–30 kHz FOC; 5 Mbit/s CAN- FD; 14.6 g	FOC current/torque driver (inner loop)
11	TowerPro MG92B servo	1–3	metal-gear micro servo; ≈ 0.29 N m (≈ 3 kg cm) at 6 V; 50 Hz PWM	Wheel brake actuator (jump-up)
Sensing				
3	mjbots MA600 breakout + ring magnet	3	TMR absolute angle; SPI 6 MHz, 3.3 V; 16-bit (65,536 counts/rev); air-gap ≤ 0.5 mm	Wheel angle: commuta- tion + state feedback
10	SparkFun BMI270 6-DoF IMU	1	3-axis accel + 3-axis gyro (no magnetometer); I ² C 400 kHz (Qwiic); ODR up to ≈ 1 kHz	Body attitude reference (tilt + rates)
Compute and Communications				
4	Teensy 4.1	1	600 MHz Cortex-M7 + FPU; na- tive CAN-FD (CAN3); 5 V supply	Flight computer: esti- mator + control law
5	CAN-FD adapter (Teensy 4.1)	1	CAN-FD transceiver, logic-level $\rightarrow$ differential; 5 Mbit/s	Bus transceiver for the MCU

Continued on next page

Table 14 (continued)

Ref	Item	Qty ^a	Key specifications	Function
6	mjbots mjcanfd-usb- 1x	1	USB ↔ CAN-FD, 5 Mbit/s	Bench interface (cali- bration, tuning)
7	JST-PH3 CAN cables	≥4	TODO: conductor gauge / current rating not in source; 3-conductor daisy-chain	CAN bus interconnect
8	Seeed XIAO ESP32-C6 + antenna	1	Wi-Fi; 3.3 V UART to MCU; TX burst ≈ 300 mA	Wi-Fi telemetry / com- mand link
9	60.4 Ω 0.5 W resistor	4	split termination, $2 \times 60.4 \Omega \approx$ 120 Ω bus impedance	CAN bus termination
17	4.7 nF capacitor	2	common-mode filter, bus midpoint to ground	CAN split-termination filter
Power				
12	Turnigy 6S LiPo (XT90)	1	22.2 V nom / 25.2 V charged; 3600 mA h; 60C; ≈ 80 W h; ≈ 600 g	Primary energy store
13	LM2596 buck module	1	step-down to 5.0 V; rated 2–3 A (derates without heatsink)	5 V logic rail
14	XT90 connector pair	1–2	rated ≈ 90 A	Battery power connec- tor
15	100 μF 16 V electrolytic	2–4	16 V part — 5 V rail only	5 V-rail bulk reservoir
19	1N5819 Schot- tky diode	1–2	1 A, 40 V	5 V-rail back-feed pro- tection
Integration				
16	100 nF 50 V capacitor	6–10	local decoupling at each IC	Per-IC supply decou- pling
18	Double-sided perfboard	1	15 cm × 9 cm, double-sided	Power / signal distribu- tion board

^a Quantity per assembled cube; motors, drivers and encoders are 3 each. A fourth actuation set (motor + moteus-n1 + MA600) is retained on the 1-DoF bench rig (S1) throughout the build, so four of each are procured.